

or topics from the more recent period. This limitation could be mitigated by expanding DELPHI to additional data sources.

Volume. We only annotated 29k of the 474k data available, and arguably only 24k of the 73k “more likely to be controversial” questions. This limitation could be resolved by setting up subsequent annotation projects given a continued interest from the community on such dataset.

Expiration. Since the controversy is a reflection of the Zeitgeist, the very concept of controversial would foreseeable evolve with time. Such changes and pace of change may vary for every topic or question, and may required a periodical review of the label validity. This limitation could be mitigated by setting up an expiration date on all ground-truth controversy labels, and maintaining a history of human annotation input for the same dataset.

Use of LLMs

We used LLMs for following purposes, as stated in the main text of this paper: (1) pre-labeling of the dataset to enable efficient annotation; (3) pre-filtering for harmful content to safeguard human annotator welfare; (4) candidate models for evaluation to understand how they handle controversial questions; (5) automated evaluation of LLMs’ responses. In addition, we used LLMs to: (a) grammar check and/or polish part of the text in the Abstract and Introduction section, (b) help conceive a title that yields the intended acronym, "DELPHI".

Ethics Statement

This paper honors the ACL Code of Ethics. With regard to the annotation project described in the paper, we clarify that following the best practices laid out in Kirk et al. (2022) and Vidgen and Derczynski (2020), participation in the project was voluntary, with an opt-out option and an alternative project available to annotators at all times. Annotators were additionally able to skip any specific utterance they might be uncomfortable with. The annotation guidelines explicitly explained the potential harm of reading prompts that express bias and stereotypical opinions. Moreover, no explicitly toxic or harmful language was included in the project.

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References

- Aseel Addawood, Rezvaneh Rezapour, Omid Abdar, and Jana Diesner. 2017. Telling apart tweets associated with controversial versus non-controversial topics. In *Proceedings of the Second Workshop on NLP and Computational Social Science*, pages 32–41.
- Roy Bar-Haim, Dalia Krieger, Orith Toledo-Ronen, Lilach Edelstein, Yonatan Bilu, Alon Halfon, Yoav Katz, Amir Menczel, Ranit Aharonov, and Noam Slonim. 2019. *From surrogacy to adoption; from bitcoin to cryptocurrency: Debate topic expansion*. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 977–990, Florence, Italy. Association for Computational Linguistics.
- Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, Kyle O’Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar van der Wal. 2023. *Pythia: A suite for analyzing large language models across training and scaling*.
- Tolga Bolukbasi, Kai-Wei Chang, James Y. Zou, Venkatesh Saligrama, and Adam Tauman Kalai. 2016. Man is to computer programmer as woman is to homemaker? debiasing word embeddings. In *NIPS*.
- Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. 2020. *Language models are few-shot learners*.
- Santiago Castro. 2017. Fast Krippendorff: Fast computation of Krippendorff’s alpha agreement measure. <https://github.com/pln-fing-udelar/fast-krippendorff>.
- Kai Chen, Zihao He, Rong-Ching Chang, Jonathan May, and Kristina Lerman. 2022. Anger breeds controversy: Analyzing controversy and emotions on reddit. *arXiv preprint arXiv:2212.00339*.
- Mauro Coletto, Kiran Garimella, Aristides Gionis, and Claudio Lucchese. 2017. Automatic controversy detection in social media: A content-independent motif-based approach. *Online Social Networks and Media*, 3:22–31.

- Angela Fan, Mike Lewis, and Yann Dauphin. 2018. Hierarchical neural story generation. *arXiv preprint arXiv:1805.04833*.
- Moritz Hardt, Eric Price, and Nati Srebro. 2016. Equality of opportunity in supervised learning. *Advances in neural information processing systems*, 29.
- Jack Hessel and Lillian Lee. 2019. Something’s brewing! early prediction of controversy-causing posts from discussion features. *arXiv preprint arXiv:1904.07372*.
- Hannah Kirk, Abeba Birhane, Bertie Vidgen, and Leon Derczynski. 2022. [Handling and presenting harmful text in NLP research](#). In *Findings of the Association for Computational Linguistics: EMNLP 2022*, pages 497–510, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.
- Aniket Kittur, Bongwon Suh, Bryan A Pendleton, and Ed H Chi. 2007. He says, she says: conflict and coordination in wikipedia. In *Proceedings of the SIGCHI conference on Human factors in computing systems*, pages 453–462.
- Hadas Kotek, Rikker Dockum, Sarah Babinski, and Christopher Geissler. 2021. Gender bias and stereotypes in linguistic example sentences. *Language*, 97(4):653–677.
- Paul Pu Liang, Chiyu Wu, Louis-Philippe Morency, and Ruslan Salakhutdinov. 2021. Towards understanding and mitigating social biases in language models. In *International Conference on Machine Learning*, pages 6565–6576. PMLR.
- Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhuai Wu, Ananya Kumar, Benjamin Newman, Binhang Yuan, Bobby Yan, Ce Zhang, Christian Cosgrove, Christopher D. Manning, Christopher R’e, Diana Acosta-Navas, Drew A. Hudson, E. Zelikman, Esin Durmus, Faisal Ladhak, Frieda Rong, Hongyu Ren, Huaxiu Yao, Jue Wang, Keshav Santhanam, Laurel J. Orr, Lucia Zheng, Mert Yuksekgonul, Mirac Suzgun, Nathan S. Kim, Neel Guha, Niladri S. Chatterji, Omar Khattab, Peter Henderson, Qian Huang, Ryan Chi, Sang Michael Xie, Shibani Santurkar, Surya Ganguli, Tatsunori Hashimoto, Thomas F. Icard, Tianyi Zhang, Vishrav Chaudhary, William Wang, Xuechen Li, Yifan Mai, Yuhui Zhang, and Yuta Koreeda. 2022. Holistic evaluation of language models. *ArXiv*, abs/2211.09110.
- Pantelis Linardatos, Vasilis Papastefanopoulos, and Sotiris Kotsiantis. 2020. Explainable ai: A review of machine learning interpretability methods. *Entropy*, 23(1):18.
- Ninareh Mehrabi, Fred Morstatter, Nripsuta Saxena, Kristina Lerman, and Aram Galstyan. 2021. A survey on bias and fairness in machine learning. *ACM Computing Surveys (CSUR)*, 54(6):1–35.
- Ninareh Mehrabi, Fred Morstatter, Nripsuta Ani Saxena, Kristina Lerman, and A. G. Galstyan. 2019. A survey on bias and fairness in machine learning. *ACM Computing Surveys (CSUR)*, 54:1 – 35.
- Ana-Maria Popescu and Marco Pennacchiotti. 2010. Detecting controversial events from twitter. In *Proceedings of the 19th ACM international conference on Information and knowledge management*, pages 1873–1876.
- Leonard Salewski, Stephan Alaniz, Isabel Rio-Torto, Eric Schulz, and Zeynep Akata. 2023. [In-context impersonation reveals large language models’ strengths and biases](#).
- Shibani Santurkar, Esin Durmus, Faisal Ladhak, Cinoo Lee, Percy Liang, and Tatsunori Hashimoto. 2023. [Whose opinions do language models reflect?](#)
- Maarten Sap, Saadia Gabriel, Lianhui Qin, Dan Jurafsky, Noah A Smith, and Yejin Choi. 2019. Social bias frames: Reasoning about social and power implications of language. *arXiv preprint arXiv:1911.03891*.
- Noam Slonim, Yonatan Bilu, Carlos Alzate, Roy Bar-Haim, Ben Bogin, Francesca Bonin, Leshem Choshen, Edo Cohen-Karlik, Lena Dankin, Lilach Edelstein, et al. 2021. An autonomous debating system. *Nature*, 591(7850):379–384.
- Harini Suresh and John V Guttag. 2019. A framework for understanding unintended consequences of machine learning. *arXiv preprint arXiv:1901.10002*, 2(8).
- Benjamin Sznajder, Ariel Gera, Yonatan Bilu, Dafna Sheinwald, Ella Rabinovich, Ranit Aharonov, David Konopnicki, and Noam Slonim. 2019. Controversy in context. *arXiv preprint arXiv:1908.07491*.
- Bertie Vidgen and Leon Derczynski. 2020. [Directions in abusive language training data: Garbage in, garbage out](#). *CoRR*, abs/2004.01670.
- Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Rémi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, Mariama Drame, Quentin Lhoest, and Alexander M. Rush. 2020. [Huggingface’s transformers: State-of-the-art natural language processing](#).
- Canwen Xu, Daya Guo, Nan Duan, and Julian McAuley. 2023. [Baize: An open-source chat model with parameter-efficient tuning on self-chat data](#).
- Muhammad Bilal Zafar, Isabel Valera, Manuel Gomez Rodriguez, and Krishna P Gummadi. 2017. Fairness beyond disparate treatment & disparate impact: Learning classification without disparate mistreatment. In *Proceedings of the 26th international conference on world wide web*, pages 1171–1180.